CHAPTER 5

The Network Layer: Control Plane

In this chapter, we'll complete our journey through the network layer by covering the **control-plane** component of the network layer—the *network-wide* logic that controls not only how a datagram is routed along an end-to-end path from the source host to the destination host, but also how network-layer components and services are configured and managed. In Section 5.2, we'll cover traditional routing algorithms for computing least cost paths in a graph; these algorithms are the basis for two widely deployed Internet routing protocols: OSPF and BGP, that we'll cover in Sections 5.3 and 5.4, respectively. As we'll see, OSPF is a routing protocol that operates within a single ISP's network. BGP is a routing protocol that serves to interconnect all of the networks in the Internet; BGP is thus often referred to as the "glue" that holds the Internet together. Traditionally, control-plane routing protocols have been implemented together with data-plane forwarding functions, monolithically, within a router. As we learned in the introduction to Chapter 4, software-defined networking (SDN) makes a clear separation between the data and control planes, implementing control-plane functions in a separate "controller" service that is distinct, and remote, from the forwarding components of the routers it controls. We'll cover SDN controllers in Section 5.5.

In Sections 5.6 and 5.7, we'll cover some of the nuts and bolts of managing an IP network: ICMP (the Internet Control Message Protocol) and SNMP (the Simple Network Management Protocol).